

🔥 XAI: Understanding Course Foundations and Motivations

FAMAF - UNC

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Disclaimer

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

Agenda

- Course Goals & Logistics
- Model Understanding: Use Cases
- Inherently Interpretable Models vs. Post-hoc Explanations
- Defining and Understanding Interpretability

Goals of this Course

Learn and improve upon the state-of-the-art literature on ML interpretability and explainability:

- Understand where, when, and why is interpretability/explainability needed
- Read, present, and discuss research papers
- Formulate, optimize, and evaluate algorithms
- Implement state-of-the-art algorithms; Do research!
- Understand, critique, and redefine literature

EMERGING FIELD!!

Course Overview

- ① **Foundations and Human Factors** - definitions, taxonomies, and cognitive aspects
- ② **Interpretability and Explainability Methods** - inherently interpretable models and post-hoc explanations
- ③ **Advanced Techniques and Evaluation** - attention-based explanations and quality metrics
- ④ **Interpretability of Large Models** - mechanistic interpretability and LLM understanding
- ⑤ **Emerging Frontiers of XAI** - automated circuit discovery and multimodal techniques

Who should take this course?

- Course particularly tailored to students interested in **research** 📝 on interpretability/explainability
 - ▶ **Not a surface level course!** 🔥
 - ▶ **Not just applications!** 🔥
- Goal is to push you to question existing work and make new contributions to the field

Class Format

- Course comprises of lectures, guests, and student presentations
- Each lecture will cover: at least 2 papers 📄 📄
- Students are **expected** to “at least” skim through the papers beforehand 🤔
- Students will divide into groups

Each breakout group is expected to come up with:

- A list of 2 to 3 weaknesses of each of the works discussed
- Strategies for addressing those weaknesses

Course Components

Research project (60%)

- 3 checkpoints (10% each) – Proposal + Baseline Implementation + Midterm Progress
- Final Report (20%)
- Final Presentation (10%)
- Teams of 2 to 3

Research Paper Presentation (30%)

- Teams of 2; Each team presents two papers in the class

Class Participation (10%)

- Being active in discussions in class
- Attending classes

Research Projects - AKA Final Exam

Requirements

- Short research paper (max 4 pages)
- Teams of 2 students
- Target: Local conference submission

Assessment

- Ongoing feedback from entire course
- Course approved when paper is submitted

Background

Understanding:

- Linear algebra
- Probability
- Algorithms
- Machine learning
- Programming in Python, Numpy, Sklearn

Familiarity with:

- Statistics
- Optimization

Motivation

[?]

Machine Learning is EVERYWHERE!!



[?]

Motivation: Why Model Understanding?

Model understanding facilitates debugging

Example: Image classification model

- Input: Dog in snow
- Prediction: Siberian Husky
- **Problem:** Model relies on snow background, not dog features
- **Solution:** Model understanding reveals incorrect feature usage

Motivation: Why Model Understanding?

Model understanding facilitates bias detection

Example: Criminal justice risk assessment

- Input: Defendant details
- Prediction: High risk to release
- **Problem:** Model uses race and gender inappropriately
- **Solution:** Model understanding reveals biased features

[Larson et. al. 2016]

Motivation: Why Model Understanding?

Model understanding helps provide recourse

Example: Loan application system

- Input: Applicant financial data
- Prediction: Loan denied
- **Solution:** Model explains actionable steps:
 - ▶ “Increase salary by \$50K”
 - ▶ “Pay credit card bills on time for next 3 months”

Motivation: Why Model Understanding?

Model understanding helps assess when to trust predictions

Example: Medical diagnosis system

- Different logic for male vs. female patients
- For females: Uses irrelevant ID number
- For males: Uses relevant symptoms
- **Decision:** Don't trust predictions for female subpopulation

Motivation: Why Model Understanding?

Summary of Use Cases

Utility	Stakeholders
Debugging	Researchers and engineers
Bias Detection	Regulatory agencies (FDA, European commission)
Recourse	End users (loan applicants)
Trust Assessment	Decision makers (doctors, judges)
Deployment Vetting	Regulatory agencies

Achieving Model Understanding

Take 1: Build inherently interpretable predictive models

Examples:

- Linear regression
- Decision trees
- Rule-based models

[Letham and Rudin 2015; Lakkaraju et. al. 2016]

Achieving Model Understanding

Take 2: Explain pre-built models in a post-hoc manner

Black Box Model → **Explainer** → Human-interpretable explanation
[Ribeiro et. al. 2016, 2018; Lakkaraju et. al. 2019]

Inherently Interpretable Models vs. Post hoc Explanations

Accuracy-Interpretability Trade-offs

In certain settings, accuracy-interpretability trade offs may exist.

Example scenarios:

- Simple boundary: Can build interpretable + accurate models
- Complex boundary: Complex models might achieve higher accuracy

[Cireşan et. al. 2012, Caruana et. al. 2006, Frosst et. al. 2017, Stewart 2020]

Inherently Interpretable Models vs. Post hoc Explanations

Sometimes, you don't have enough data to build your model from scratch.

And, all you have is a (proprietary) black box!

[Ribeiro et. al. 2016]

Inherently Interpretable Models vs. Post hoc Explanations

Recommendation

**If you can build an interpretable model which is also adequately accurate for your setting, DO IT!
Otherwise, post hoc explanations come to the rescue!**

Defining and Understanding Interpretability

Motivation for Interpretability

ML systems are being deployed in complex high-stakes settings:

- Accuracy alone is no longer enough
- Auxiliary criteria are important:
 - ▶ Safety
 - ▶ Nondiscrimination
 - ▶ Right to explanation

Motivation for Interpretability (cont.)

Auxiliary criteria are often hard to quantify (completely):

- E.g.: Impossible to enumerate all scenarios violating safety of an autonomous car

Fallback option: interpretability

- If the system can explain its reasoning, we can verify if that reasoning is sound w.r.t. auxiliary criteria

Prior Work: Defining and Measuring Interpretability

Little consensus on what interpretability is and how to evaluate it.

Interpretability evaluation typically falls into:

① Evaluate in the context of an application

- ▶ If a system is useful in a practical application or a simplified version, it must be interpretable

② Evaluate via a quantifiable proxy

- ▶ Claim some model class is interpretable and present algorithms to optimize within that class
- ▶ E.g. rule lists

“You will know it when you see it!”

Lack of Rigor?

Yes and No

- Previous notions are reasonable
- **However:**
 - ▶ Are all models in all “interpretable” model classes equally interpretable?
 - ▶ Model sparsity allows for comparison
 - ▶ How to compare a linear model with a decision tree?
 - ▶ Do all applications have same interpretability needs?

Important to formalize these notions!!!

What is Interpretability?

Definition: Ability to explain or to present in understandable terms to a human

No clear answers in psychology to:

- What constitutes an explanation?
- What makes some explanations better than the others?
- When are explanations sought?

When and Why Interpretability?

Not all ML systems require interpretability

- E.g., ad servers, postal code sorting
- No human intervention
- **No explanation needed because:**
 - ▶ No consequences for unacceptable results
 - ▶ Problem is well studied and validated well in real-world applications → trust system's decision

When do we need explanation then?

When and Why Interpretability?

Incompleteness in problem formalization

- Hinders optimization and evaluation
- **Incompleteness** $\neq$ **Uncertainty**
- Uncertainty can be quantified
- E.g., trying to learn from a small dataset (uncertainty)

Incompleteness: Illustrative Examples

Scientific Knowledge

- E.g., understanding the characteristics of a large dataset
- Goal is abstract

Safety

- End to end system is never completely testable
- Not possible to check all possible inputs

Ethics

- Guard against certain kinds of discrimination which are too abstract to be encoded
- No idea about the nature of discrimination beforehand

Taxonomy of Interpretability Evaluation

	Humans	Tasks
Application-grounded Evaluation	Real Humans	Real Tasks
Human-grounded Evaluation	Real Humans	Simple Tasks
Functionally-grounded Evaluation	No Real Humans	Proxy Tasks

Claim of the research should match the type of the evaluation!

Application-grounded evaluation

Real humans (domain experts), real tasks

- Domain experts experiment with exact application task
- Domain experts experiment with a simpler or partial task
 - ▶ Shorten experiment time
 - ▶ Increases number of potential subjects
- Typical in HCI and visualization communities

Human-grounded evaluation

Real humans, simplified tasks

- Can be completed with lay humans
- Larger pool, less expensive

Potential experiments:

- Pairwise comparisons
- Simulate the model output
- What changes should be made to input to change the output?

Functionally-grounded evaluation

No humans, just proxies

Appropriate for:

- A class of models already validated (E.g., decision trees)
- A method is not yet mature
- Human subject experiments are unethical

What proxies to use?

Potential experiments:

- Complexity (of a decision tree) compared to other models of the same (similar) class
- How many levels? How many rules?

Open Problems: Design Issues

- What proxies are best for what real world applications?
- What factors to consider when designing simpler tasks in place of real world tasks?

Taxonomy based on applications/tasks

Global vs. Local

- High level patterns vs. specific decisions

Degree of Incompleteness

- What part of the problem is incomplete? How incomplete is it?
- Incomplete inputs or constraints or costs?

Time Constraints

- How much time can the user spend to understand explanation?

Taxonomy based on applications/tasks (cont.)

Nature of User Expertise

- How experienced is end user?
- Experience affects how users process information
- E.g., domain experts can handle detailed, complex explanations compared to opaque, smaller ones

Note: These taxonomies are constructed based on intuition and are not data or evidence driven. They must be treated as hypotheses.

Taxonomy based on methods

Basic units of explanation:

- Raw features? E.g., pixel values
- Semantically meaningful? E.g., objects in an image
- Prototypes?

Number of basic units of explanation:

- How many does the explanation contain?
- How do various types of basic units interact?
- E.g., prototype vs. feature

Taxonomy based on methods (cont.)

Level of compositionality:

- Are the basic units organized in a structured way?
- How do the basic units compose to form higher order units?

Interactions between basic units:

- Combined in linear or non-linear ways?
- Are some combinations easier to understand?

Uncertainty:

- What kind of uncertainty is captured by the methods?
- How easy is it for humans to process uncertainty?

Relevant Conferences to Explore

ML/AI Venues:

- ICML, NeurIPS, ICLR
- UAI, AISTATS, KDD, AAAI

Ethics/HCI Venues:

- FAccT, AIES
- CHI, CSCW, HCOMP

Breakout Groups

Say hi to your neighbors! Introduce yourselves!

- What topics are you most excited about learning as part of this course?
- Are you convinced that model interpretability/explainability is important?
- Do you think we can really interpret/explain models (correctly)?
- What is your take on inherently interpretable models vs. post hoc explanations? Would you favor one over the other? Why?

References I